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Task 3 — Docker Storage

1. Container-Layer Data Loss

```
PS C:\Users\HP> docker run -it --name storage-test ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
f3db1cd94078: Pull complete
06e9d71331fb: Pull complete
f79afc4bbefe: Download complete
Digest: sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b
Status: Downloaded newer image for ubuntu:latest
root@8197e0c185d4:/#
```

```
C:\Users\HP>docker rm -f storage-test
storage-test
```

```
C:\Users\HP>docker run -it --name storage-test ubuntu bash
root@a01b6d288c1b:/#
root@a01b6d288c1b:/# echo "This data will be lost" > /data.txt
root@a01b6d288c1b:/#
```

```
C:\Users\HP>docker run -it --name storage-test ubuntu bash
root@a01b6d288c1b:/#
root@a01b6d288c1b:/# echo "This data will be lost" > /data.txt
root@a01b6d288c1b:/# cat /data.txt
This data will be lost
```

```
C:\Users\HP>docker run -it --name storage-test ubuntu bash
root@a01b6d288c1b:/#
root@a01b6d288c1b:/# echo "This data will be lost" > /data.txt
root@a01b6d288c1b:/# cat /data.txt
This data will be lost
root@a01b6d288c1b:/# exit
exit
```

2. Permanent Storage Using Docker Named Volume

```
C:\Users\HP>docker run --rm ubuntu cat /data.txt
cat: /data.txt: No such file or directory

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
```

```
C:\Users\HP>docker volume create app-data
app-data

C:\Users\HP>docker volume ls
DRIVER    VOLUME NAME
local     81fe42e81eb022af893300de6bff715fc7c95d2dedeff98105f6bb181186c3f5
local     app-data
```

```
C:\Users\HP>docker run -it --name data-container --mount source=app-data,target=/data ubuntu bash
root@355020210a7f:/#
```

```
C:\Users\HP>docker run -it --name data-container --mount source=app-data,target=/data ubuntu bash
root@355020210a7f:/# echo "Hello from Docker Volume" > /data/message.txt
root@355020210a7f:/# cat /data/message.txt
Hello from Docker Volume
```

```
C:\Users\HP>docker volume create app-data
app-data

C:\Users\HP>docker volume ls
DRIVER    VOLUME NAME
local     81fe42e81eb022af893300de6bff715fc7c95d2dedeff98105f6bb181186c3f5
local     app-data
```

3. Permanent Storage Using Bind Mount

```
C:\Users\HP>mkdir "$HOME\docker-storage"

C:\Users\HP>dir "$HOME\docker-storage"
 Volume in drive C is Windows
 Volume Serial Number is 2C9B-CD6A

 Directory of C:\Users\HP\$HOME\docker-storage

08-09-2026  20:03    <DIR>          .
08-09-2026  20:03    <DIR>          ..
               0 File(s)              0 bytes
               2 Dir(s)  245,146,419,200 bytes free

C:\Users\HP>docker run -it --name bind-container --mount type=bind,source="$HOME\docker-storage",target=/data ubuntu bas
h

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: $HOME\docker-storage%!(EXTRA string=is not a valid Windows path)

Run 'docker run --help' for more information
```

```
C:\Users\HP>mkdir "C:\Users\HP\docker-storage"

C:\Users\HP>dir "C:\Users\HP\docker-storage"
 Volume in drive C is Windows
 Volume Serial Number is 2C9B-CD6A

 Directory of C:\Users\HP\docker-storage

08-09-2026  20:05    <DIR>          .
08-09-2026  20:05    <DIR>          ..
               0 File(s)              0 bytes
               2 Dir(s)  245,076,037,632 bytes free

C:\Users\HP>docker run -it --name bind-container --mount type=bind,source="C:\Users\HP\docker-storage",target=/data ubun
tu bash
root@38e1da3f9e37:/#
```

```
C:\Users\HP>docker run -it --name bind-container --mount type=bind,source="C:\Users\HP\docker-storage",target=/data ubun
tu bash
root@38e1da3f9e37:/# echo "Hello from Docker Bind Mount" > /data/message.txt
root@38e1da3f9e37:/# cat /data/message.txt
Hello from Docker Bind Mount
```

```
C:\Users\HP>docker rm bind-container
bind-container

C:\Users\HP>type "C:\Users\HP\docker-storage\message.txt"
Hello from Docker Bind Mount
```